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Upgrade Guide for SAP S/4HANA

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Document History

Version	Date	Description
1.0	October 11, 2023	Version for SAP S/4HANA 2023.
2.0	February 28, 2024	Version for SAP S/4HANA 2023 FPS01.
2.1	July 08, 2024	Minor correction.
3.0	October 08, 2024	Version for SAP S/4HANA 2023 FPS02.
4.0	February 26, 2025	Version for SAP S/4HANA 2023 FPS03.
5.0	August 6, 2025	Version for SAP S/4HANA 2023 SPS04.

1 Upgrade Guide for SAP S/4HANA

This guide explains the upgrade process for the upgrade from all lower releases of SAP S/4HANA to SAP S/4HANA 2023 (any feature package or support package stack).

2 Getting Started

Read this guide carefully to get an overview of how to do the upgrade from a lower release of SAP S/4HANA to SAP S/4HANA 2023 (any feature package or support package stack) .

This guide has to be used together with the following documents and tools:

- Simplification Item Catalog
  This tool allows you to search and browse upgrade-relevant simplification items online.
- *Software Update Manager 2.0 <latest version>*
- *Maintenance Planner User Guide*
- SAP Readiness Check for SAP S/4HANA upgrades
  This tool analyzes your SAP S/4HANA source system and highlights important aspects of the upgrade to your SAP S/4HANA target system, such as identification of relevant simplification items, high-level custom code analysis, add-on compatibility, and more.

  > → Recommendation
  >
  > Although not mandatory, this tool is highly recommended. It is relevant for product release upgrades (for example from SAP S/4HANA 2022 to 2023), but not for updates from one FPS or SPS to the next (for example from SAP S/4HANA 2022 FPS02 to SPS03).
- *Custom Code Migration Guide for SAP S/4HANA*

You find more information about these tools and guides (including where to find them), as well as additional important documents and SAP Notes relevant for the upgrade in the section Documentation and SAP Notes for the Upgrade [page 5].

The section Overview of the Upgrade Process [page 8] provides information about the different phases of the upgrade and the tools involved.

The sections Preparing the Upgrade [page 14] and Realizing the Upgrade [page 34] provide details for those upgrade phases.

2.1 Documentation, Tools, and SAP Notes for the Upgrade

Required Documents, Tools, and SAP Notes

You require at least the following documents and SAP Notes.

Document	Available at	Comment
Simplification Item Catalog This tool enhances and replaces the *Simplification List for SAP S/4HANA* (PDF). The PDF is still available, but we recommend using the catalog.	https://help.sap.com/s4hana_op_2023 ▶ *Implement* ▶ *Conversions & Upgrade Assets* ▶	Lists the simplifications in comparison to the previous releases, SAP S/4HANA 1511, 1610, 1709, 1809, 1909, 2020, 2021, 2022 and 2023 (all feature package stacks), such as, simplified functions, merged database tables, and new data models.
Software Update Manager 2.0 <latest SP>	http://help.sap.com/sltoolset ▶ *Software Logistics Toolset (SL Toolset)* ▶ *System Maintenance* ▶ *System Maintenance Scenarios* ▶ *Software Update/Upgrade using SUM* ▶	Describes how to do the upgrade to SAP S/4HANA 2023 using the Software Update Manager tool.
Maintenance Planner User Guide	http://help.sap.com/maintenanceplanner	Describes how to use the Maintenance Planner for calculating and downloading the required stack.xml file and software packages.
SAP Readiness Check for SAP S/4HANA upgrades	https://sap.com/readinesscheck	This tool analyzes your SAP S/4HANA source system and highlights important aspects of the upgrade to your SAP S/4HANA target system, such as identification of relevant simplification items, high-level custom code analysis, add-on compatibility, and more. → Recommendation Although not mandatory, this tool is highly recommended. It is relevant for product release upgrades (for example from SAP S/4HANA 2022 to 2023), but not for updates from one FPS or SPS to the next (for example from SAP S/4HANA 2022 FPS02 to SPS03).
Custom Code Migration Guide for SAP S/4HANA	https://help.sap.com/s4hana_op_2023 ▶ *Implement* ▶ *Conversion & Upgrade Assets* ▶	Provides instructions on how to adapt your custom code.
SAP S/4HANA 2023: Release Information Note	SAP Note 3307222	-

Document	Available at	Comment
SAP S/4HANA 2023: Feature Package Stack 00: Additional Release Information	SAP Note 3351047	-
SAP S/4HANA 2023: Restriction Note	SAP Note 3348949	-
SAP S/4HANA Add-on Note	SAP Note 2214409	-
Notes relevant for the Simplification Item Check	SAP Notes 2399707 and 2502552	With these notes, the SI-Check is delivered.
SAP S/4HANA 2023 - application specific notes in system conversion / upgrade preparation phase	SAP Note 3340515	Information about errors that may occur during the preparation phase of the upgrade.
Additional information about the upgrade with the Software Upgrade Manager tool.	SAP Note for the latest version of SUM 2.0 at https://support.sap.com/sltoolset ▶ *System Maintenance* ▶ *System Maintenance Scenarios* ▶ *Software Update/Upgrade using SUM* ▶	Information about errors that may occur during the SUM phase of the upgrade.
SAP S/4HANA 2023 - application specific notes in system conversion / upgrade follow-on phase	SAP Note 3340526	Information about errors that may occur during the follow-on phase of the upgrade.
SAP S/4HANA Foundation 2023: Release Information Note	SAP Note 3306766	-
SAP Fiori for SAP S/4HANA 2023: Release Information Note	SAP Note 3336823	-
Maintenance and Update Strategy for SAP Fiori Front-End Server	SAP Note 2217489	Check dependencies according to the SAP Fiori front-end server maintenance and update strategy
General Information: Installation details of SAP Fiori front-end server 2023 for SAP S/4HANA	SAP Note 3348673	-
ABAP Platform 2023 – General Information	SAP Note 3346290	-
Java components for SAP S/4HANA 2023 – Restrictions	SAP Note 3247591	-

Additional Information and SAP Notes

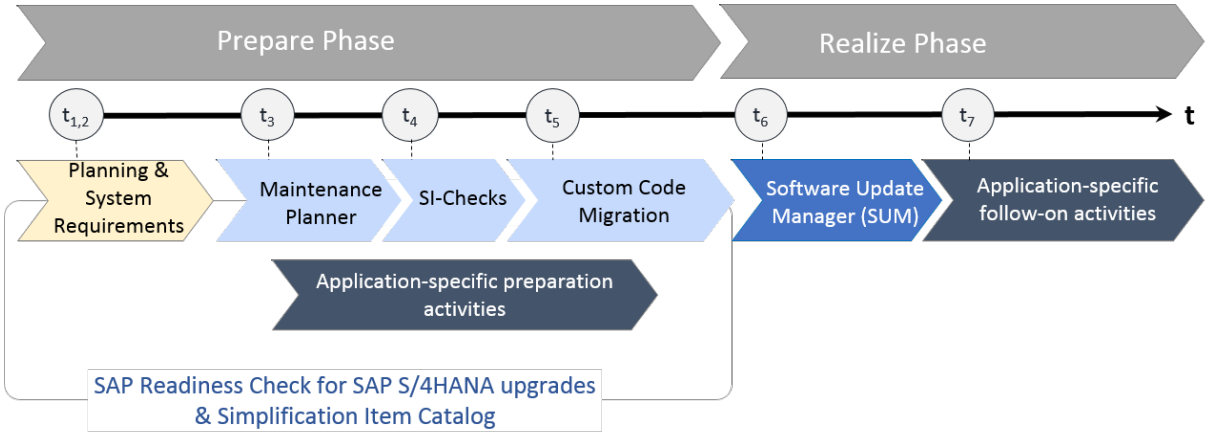
The following table lists important additional documents and SAP Notes.

Document	Available at	Comment
SAP S/4HANA: Always-Off Business Functions	SAP Note 2240359	-
SAP S/4HANA: Always-On Business Functions	SAP Note 2240360	-
Uninstalling ABAP Add-ons	SAP Note 2011192	-
SPAU Adjustment	SAP Note 2344014	-
Automated guided steps for enabling Note Assistant for TCI and Digitally Signed SAP Notes	SAP Note 2836302	-
SAP S/4HANA System Conversion / Upgrade: Measures to reduce technical downtime	SAP Note 2351294	-
Superfluous field "BUFFERED" causes exception in Consolidation and Mass Processing and potential S/4 upgrade error	SAP Note 2781530	-
SAP S/4HANA Service: Upgrade Information for WebClient UI Framework	SAP Note 2834255	-

2.2 Overview of the Upgrade Process

The overview of the upgrade process shown in the figure below (including the tools, the phases, and the activities involved in the upgrade process) is intended to help you to plan and perform the upgrade from lower SAP S/4HANA releases to higher SAP S/4HANA releases.

> → Recommendation
>
> We recommend that you do the activities in the sequence shown in the figure and explained in the sections below.



Upgrade to SAP S/4HANA: Sequence

Prepare Phase

Simplification Item Catalog

To enable you to do an optimal planning of your upgrade, we provide the Simplification Item Catalog. The Simplification Item Catalog is a tool that allows you to search or browse the complete collection of simplification items grouped by product version and presented by application or functional area. Each simplification item details the steps that need to be taken for the conversion and/or the upgrade from a business and a technical point of view, both preparatory and follow-on steps and each item is available as a SAP Note. You need to search for the upgrade-relevant simplification items – most of the simplification items are not relevant for the upgrade.

> → Recommendation
>
> We recommend that you consume the simplification items also via the SAP Readiness Check for SAP S/4HANA upgrades. Based on an analyis of the configuration, data, and usage of your productive SAP S/4HANA source release, the SAP Readiness Check for SAP S/4HANA for upgrades identifies the simplification items relevant for your upgrade to your SAP S/4HANA target release.
>
> For more information about how to use the SAP Readiness Check for SAP S/4HANA upgrades, see https://sap.com/readinesscheck.

You find the Simplification Item Catalog at https://launchpad.support.sap.com/#/sic/.

> ⓘ Note
>
> When using the Simplification Item Catalog directly instead of the SAP Readiness Check for SAP S/4HANA upgrades, please note that the catalog contains **all** the simplification items, not only those that are relevant for your upgrade project. This means you need to identify those which are actually relevant for your upgrade (which the SAP Readiness Check for SAP S/4HANA upgrades does automatically). And when you have done this, you also need to look at the relevant simplification SAP Notes themselves to get all the available information. The SAP Notes contain all the required information for each simplification item. The SAP Readiness Check for SAP S/4HANA upgrades links directly to the most recent version of the SAP Notes.

The simplification items are integrated with the Simplification Item-Check and custom code migration tools. These tools provide detailed reports when the code of your lower SAP S/4HANA release does not comply with the scope and data structures of the SAP S/4HANA release to which you are upgrading.

Before starting the realization phase of the upgrade process, you have to do the following preparatory activities:

1. **Planning**

   > → Recommendation
   >
   > We highly recommend to run the SAP Readiness Check for SAP S/4HANA upgrades in advance of your upgrade project to identify any issues you need to consider and activities you need to do in preparation for your project.

   For more information about how to use the SAP Readiness Check for SAP S/4HANA upgrades, see https://sap.com/readinesscheck.
   See also, SAP Readiness Check Versus SI-Check [page 11].
2. **System Requirements**
   You need to be aware of system requirements, see Supported Start Releases [page 15].
3. **Maintenance Planner**
   You need to run the maintenance planner tool as a first step in the upgrade process. It checks your components, add-ons, and business functions to ensure compatibility with the SAP S/4HANA release to which you are upgrading and also creates the stack file used during the upgrade process (done by the Software Update Manager tool). For more information, see Maintenance Planner [page 15].

   > ⓘ Note
   >
   > This step is mandatory, because the Software Update Manager requires the stack file for the upgrade process.
   >
   > The Maintenance Planner has replaced the Maintenance Optimizer, which is not supported by SAP S/4HANA.

4. **Simplification Item-Check**
   This check identifies the steps you need to take to make sure that your system can technically be upgraded and that your business processes can start running directly after the upgrade process has been completed. The Simplification Item-Check (SI-Check) is called by the Software Update Manager. For more information, see Simplification Item-Check [page 17].

   > ⓘ Note
   >
   > This step is mandatory and will be triggered by the Software Update Manager. Ideally, you run the SI-Check report early in the upgrade process so that you can take note of the preparation information provided by the SI-Check results (see also SAP Note 2399707).

   See also, SAP Readiness Check Versus SI-Check [page 11].
5. **Custom Code Migration**
   The custom code migration tool checks your custom code against the simplification items relevant for the upgrade. For more information, see Custom Code Analysis [page 19].

> **Note**
>
> While not mandatory, this step is highly recommended. Ideally, you combine the upgrade project with housekeeping activities for your existing custom code base. In particular, you need a consolidated view of productively used custom developments and you should remove custom code that is no longer used.
>
> For more information, see the Custom Code Migration Guide for SAP S/4HANA at https://help.sap.com/s4hana_op_2023 ▶ *Implement* ▶ *Conversion & Upgrade Assets* ▶.

6. **Application-specific preparation activities**
   You may need to do application-specific manual preparation activities. These activities are identified by the SI-Checks and custom code migration checks and documented by SAP Notes, which you get through the check results.

> **→ Recommendation**
>
> We recommend that you do the steps in the prepare phase in the sequence listed above. It is, however, technically possible to do them independently or in parallel.

Realize Phase

After you have done the steps of the prepare phase, you continue with the activities of the realize phase:

1. **Software Update Manager (SUM)**
   When you have completed the steps above, and have implemented all the adaptations required to ensure your system and your custom code is suited to the SAP S/4HANA release to which you are upgrading, you then run the SUM. The SUM does the actual software upgrade. For more information, see Realizing the Upgrade [page 34].
   Migration of application data usually happens during release upgrades or updates while the system is down. SAP provides the Silent Data Migration Infrastructure (SDMI) to allow the migration of application data after the upgrade during business operations. For more information, see Silent Data Migration [page 34].
2. **Application-specific follow-on activities**
   You may need to do application-specific manual activities. These activities are identified by the SI-Check and custom code migration checks and documented by SAP Notes, which you get through the checks.

2.3 SAP Readiness Check Versus SI-Check

The tools SAP Readiness Check SAP S/4HANA upgrades and Simplification Item-Check (SI-Check) are used to support the upgrade process, see also Overview of the Upgrade Process [page 8]. The following table explains the differences between the two tools.

Tool Comparison

	SAP Readiness Check	Simplification Item-Check /SDF/RC_START_CHECK
Type	Customer self-service provided by SAP Digital Business Services (included as standard support)	Standalone report
SAP Solution Manager	Not required	Not required
User Interface	Cloud-based web interface	SAP GUI
Included Checks	• Simplification item relevance • Custom code • Recommended SAP Fiori apps • Add-on compatibility • SAP Custom Development projects • Check of system consistency before conversion or upgrade • Integration analysis	• Simplification item relevance • Check of system consistency before conversion or upgrade
Called by Software Update Manager (SUM)	No	Yes
Mandatory	No	Yes
Use Cases	• Conversion to SAP S/4HANA • Upgrade to a higher SAP S/4HANA release or Feature Pack Stack	• Conversion to SAP S/4HANA • Upgrade to a higher SAP S/4HANA release or Feature Pack Stack

While both the SAP Readiness Check and the Simplification Item-Check can display information on simplification item relevance and consistency, they serve different purposes and are both required:

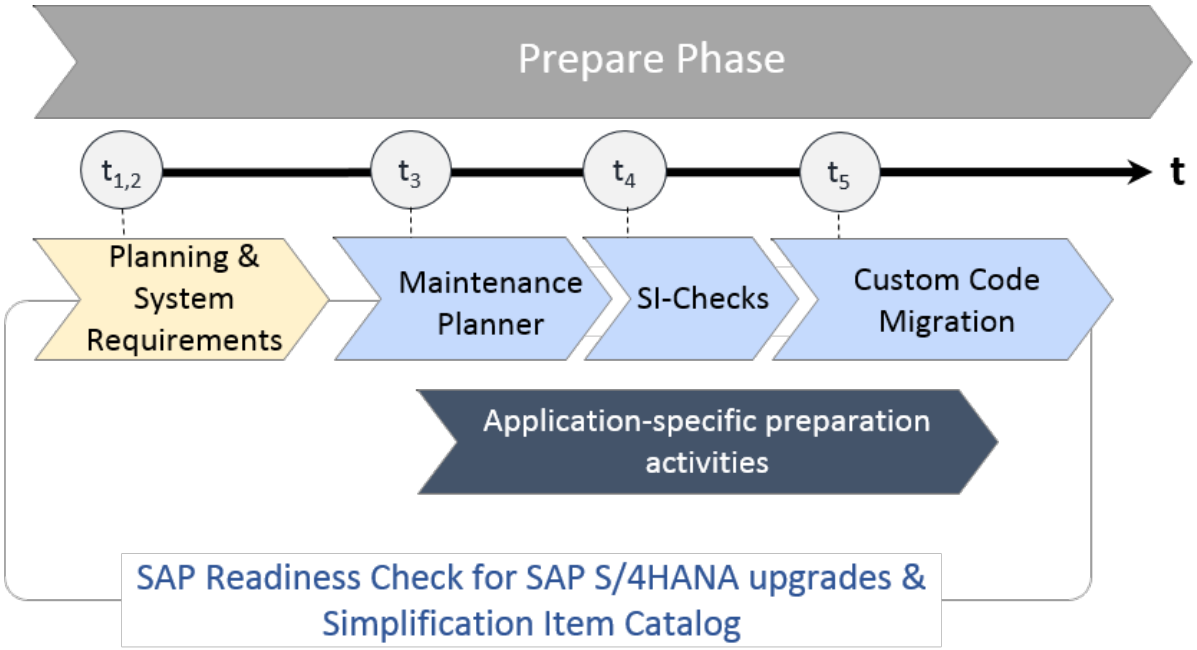
- SAP Readiness Check for SAP S/4HANA upgrades is mainly a planning tool which you run early in an upgrade project to get an overview about the required activities – including tasks related to simplification items. It is also useful for tracking these activities during the project, for example, using the burndown chart that shows simplification item consistency errors.
- The Simplification Item-Check is used on operational level during the cleanup of the simplification item consistency errors, up to the downtime, before which the SUM tool does the final executions of the SI-Check. So, the SI-Check offers a more detailed control over running the checks. This includes the option to re-run checks for individual simplification items instead of running all checks like SAP Readiness Check for SAP S/4HANA upgrades does. This option can significantly reduce the check runtime when you just want to re-check if a specific error which you have just fixed is really gone.
  In addition, the SI-Check offers the option to exempt certain types of errors (for details, see the blog post linked below).

For more information, see:

- SAP Note 3059197 (about the SAP Readiness Check).
- In this guide Simplification Item-Check [page 17]

- Blog post https://blogs.sap.com/2018/03/26/sap-s4hana-simplification-item-check-how-to-do-it-right./

3 Preparing the Upgrade



Upgrade to SAP S/4HANA: Preparations

For the upgrade to SAP S/4HANA you have to prepare an overall project plan and schedule the tasks. The preparation activities described in this section give you an idea of what is involved. Please make sure that you also check the upgrade-relevant simplification items. For a short overview of the complete process, see Overview of the Upgrade Process [page 8].

> ### Note
>
> In addition, review the planning section of the Software Update Manager guide *Software Update Manager (SUM) 2.0 <latest version>* at https://support.sap.com/sltoolset *Software Logistics Toolset* *System Maintenance* *System Maintenance Scenarios* *Softwar* *e Update/Upgrade using SUM* .

> ### Note
>
> You find some preparation steps in this guide and you get additional ones with the Simplification Item-Check and custom-code checks as well as with the Simplification Item Catalog or the SAP Readiness Check for SAP S/4HANA upgrades.
>
> Please see also SAP Note 3340515.

3.1 Supported Start Releases

Please check the *SAP S/4HANA 2023: Release Information Note* 3307222 for the required feature package stack equivalencies for the upgrade from SAP S/4HANA 1709, 1809, 1909, 2020, 2021, or 2022 (all feature package and support package stacks) to 2023 (all feature package and support package stacks).

If you intend to move from an SAP ERP release to SAP S/4HANA 2023, you need to do a system conversion, not an upgrade. For information about system conversion, see the *Conversion Guide for SAP S/4HANA* at https://help.sap.com/s4hana_op_2023 *Implement* *Guides* .

3.2 Maintenance Planner

Overview

You have to use the Maintenance Planner to do the upgrade to an SAP S/4HANA release. The Maintenance Planner generates the download files (add-ons, packages, DBDs, and the stack configuration file) that the Software Update Manager (SUM) uses to do the upgrade. In particular, the Maintenance Planner checks if the following items are supported for the upgrade:

- Add-ons to your system
- Active business functions in your system
- Industry solutions

If there is no valid upgrade path for any of the items listed above (for example, an add-on is not released for the upgrade yet), the Maintenance Planner prevents the upgrade. After the check, the Maintenance Planner creates the stack configuration file (stack.xml).

> **Note**
>
> You need to run the Maintenance Planner before the Simplification Item-Check (SI-Check), because the SI-Check report also requires the stack.xml file as an input.

For more information, see the Maintenance Planner User Guide at http://help.sap.com/maintenanceplanner.

Preparing the Use of the Maintenance Planner

Besides the general preparation described in the *Maintenance Planner User Guide*, you have to make the following preparatory steps specific to the upgrade to SAP S/4HANA:

1. Set up the Maintenance Planner as described in the *Maintenance Planner User Guide* available at http://help.sap.com/maintenanceplanner.
2. Check the SPAM/SAINT version in your source release system.

   > **Note**
   >
   > SPAM/SAINT patch 71 (or higher) is required.

3. Apply SAP Note 2186164 to your SAP Solution Manager system.
4. Create an RFC-connection between your source release system and SAP Solution Manager:
    1. In SAP Solution Manager, choose *SOLMAN_WORKCENTER* > *Solution Manager Administration* > *Landscape (Technical Systems)*.
    2. Click *RFC connections* > *Start New Window*.
    3. Select client 000 and choose *Advanced Mode*.
    4. Under *Export Options*, fill in the field *SAP Solution Manager Host Name*.
    5. Under *Communication RFCs*, select *Create SM_<SID>CLNT<CLNT>_READ*.
    6. Select *Create New User*.
    7. Enter a password.
    8. Enter a user and password under *Managed System Setup Administrator*, and choose *Test Login* in the group *Administration Users* to check that it works.

Business Functions

Business functions can have the following status: `always_on`, `customer_switchable`, and `always_off`. This results in the following behavior during the upgrade:

- If a business function was switched on in the start release system, but defined as `always_off` in the SAP S/4HANA target release, then an upgrade is not possible with this release.
- If a business function was switched off in the start release system, but defined as `always_on` in the SAP S/4HANA target release, then the business function will be activated during the upgrade.
- If a business function is defined as `customer_switchable` in the SAP S/4HANA target release, it will keep the state defined in the target release during the upgrade.

For more information about always-off and always-on business functions in SAP S/4HANA, see SAP Notes 2240359 and 2240360.

Add-Ons

For a list of supported add-ons, see SAP Note 2214409.

For information about uninstalling add-ons, see SAP Note 2011192.

Industry Solutions

For information about supported industry solutions, see SAP Note 3348949.

3.3 System Requirements

Before upgrading your SAP software you must make sure that the required infrastructure components such as SAP HANA database and operating system fullfill the minimum technical requirements. You find the respective information in the SAP Product Availability Matrix (PAM). For more information, see https://apps.support.sap.com/sap/support/pam.

3.4 Simplification Item-Check

The Simplification Item-Check (SI-Check) is called by the Software Update Manager (SUM) tool and it identifies the simplification items relevant for your system upgrade.

The SI-Check is delivered with SAP Notes 2399707 and 2502552. SAP Note 2399707 delivers the new check report; SAP Note 2502552 delivers the check classes via transport-based correction instructions (TCI) and prerequisite notes.

> **Note**
>
> If you follow the process described in SAP Note 2502552 exactly, you do not need to read any of the SAP Notes required by SAP Note 2502552. They do not contain any manual tasks and are independent of business functionality. After you run the SI-Check, the result list will contain the SAP Notes you have to look at in detail.

Implementing TCI and the SI-Check

1. To enable TCI implementation, follow the instructions provided in SAP Note 2187425 (see especially the PDF file attached to the SAP Note).

   > **Note**
   >
   > If you do not enable your system to install TCI notes, SAP Note 2502552 will be implemented without the TCI part, which will cause missing check class errors. However, if your system is on SAP NetWeaver / ABAP Platform releases listed in the table, TCI is already enabled:
   >
   > TCI Enabled Releases
   >
   > | Release | Minimum Support Package |
   > |---|---|
   > | 750 | SP05 |
   > | 751 | SP02 |
   > | 752 | SP00 |

Release	Minimum Support Package
753	SP00
Or higher	-

2. Follow the process description in SAP Note 2502552 to implement the SI-Checks.

Running the SI-Check

You run the SI-Check to identify the simplification items relevant for your upgrade project. We recommend to do this early in the project, to get an overview of the upgrade scope.

1. Start the report `/SDF/RC_START_CHECK` in transaction SA38.
2. In the section *Simplification Item Check Options*, choose the target SAP S/4HANA version.
3. Choose the mode in which you want to run the checks:
   - In online mode: The results are displayed immediately after the check is finished.
   - As a background job: We recommend this option if the check will need a long running time.
4. Run the checks to get an overview over all simplification items and then check the system consistency with *Check Consistency for All*.
5. Check the results.

The report returns a list of relevant and irrelevant simplification items. We recommend that you check every relevant simplification item for the impact it will have on your upgrade project.

Some simplification items have a consistency check. The consistency check identifies inconsistencies in the system that would be a problem during the SUM process. It also provides additional information on how to solve the problem. Some simplification items do not have a consistency check, but nevertheless are relevant. This means that from a technical perspective an upgrade of your system is possible without any action from you, but there will be an impact and you should investigate it.

Some of the error messages from the consistency check aim for you to read and understand the information provided by the consistency check error – for example, the information might be about data loss incurred during the upgrade process. These errors can only be processed by creating an exception for them which is logged in the system. The errors are highlighted in the report in the column *Exemption Possible*, so that you can see which errors need an exemption.

For more information about how to use the SI-Checks, see the user guide in SAP Note 2399707.

SI-Check Messages and Their Meanings

The SI-Checks checks your system for data consistency (necessary for the upgrade of existing data) and effects on the data after the upgrade. The severity of the resulting messages indicates whether there are no

inconsistencies, warnings you should look at, or inconsistencies you need to resolve. The results are provided as green, yellow, or red messages:

Message Overview

Message Color	Message Return Code	Descriptions
Green	0	These messages indicate that there are no technical or other inconsistencies. They also function as status messages and indicate that the upgrade can go ahead.
Yellow	4	These messages provide important information (or warnings) about various topics, such as changes to business functions, or adaptations that can be done either before or after the upgrade (but must be done to ensure business continuity). They indicate that there are no technical problems for the upgrade itself. However, some of these yellow messages provide information about possible data changes, changes in functionality, or required tasks after the upgrade. For instance, per default, no custom table appends are retained during the upgrade – the data from those fields is just lost. **⚠ Caution** Please do not underestimate yellow messages. Read them carefully and do what is required to ensure system consistency after the upgrade, so that business continuity is not compromised.
Red	7	These messages show skippable errors that indicate important changes for the affected simplification items. The errors are no inconsistencies, but you need to confirm that you have read and understood them. Usually they point out imminent data loss, deprecation, or required activities after the upgrade. Once you have confirmed (that is, exempted) them, they will no longer act as blockers for the upgrade.
	8 or 12	These messages always indicate inconsistencies in the system. These inconsistencies need to be solved before the upgrade, as they would act as blockers in later SUM phases. Please read the messages for information about how you can solve the inconsistencies. **ⓘ Note** The Software Update Manager (SUM) runs the SI-Check at the beginning of the SUM process and again just before the downtime. At the beginning, the SUM process is stopped if a message with return code 12 is detected. Just before the downtime, the SUM process is stopped if a message with return code 8 or higher is found. So, for the SUM process to run successfully, you need to make sure that all the inconstistencies found by the SI-Check with message return codes at 8 or higher are solved.

3.5 Custom Code Analysis

The custom code migration checks are based on the simplification item concept. Before upgrading to SAP S/4HANA 2023 (or any of its feature package or support package stacks), you need to check your custom code

against the SAP S/4HANA simplifications in an SAP S/4HANA 2023 system (on the correct feature package or support package stack) with the Custom Code Migration tool or the ABAP test cockpit (ATC). These tools provide you with a list of instances where your custom code does not comply with the scope and data structure of SAP S/4HANA 2023 (any feature package or support package stack).

> **Note**
>
> Although it is not mandatory, we highly recommend that you perform these checks regularly so that your code remains in compliance with the SAP S/4HANA code.

For additional information about the Custom Code Migration tool, see:

- SAP Note 2241080 for information about how to download the simplification database.
- *Custom Code Migration Guide for SAP S/4HANA* at https://help.sap.com/s4hana_op_2023 *Implement Conversion & Upgrade Assets*.

3.5.1 Custom Code Analysis with Baseline

The following sections describe how to use the baseline functionality in the ABAP test cockpit and the Custom Code Migration app to filter legacy findings in the context of an SAP S/4HANA upgrade:

- Preparing the Custom Code Analysis [page 20]
- Using the ABAP Test Cockpit [page 22]
- Using the Custom Code Migration App [page 28]

3.5.1.1 Preparing the Custom Code Analysis

To prepare the custom code analysis, you need to perform the following tasks:

1. Configure users
2. Apply SAP Notes
3. Implement the Custom Code Migration app
4. Import the Simplification Database

Configuring the Users

In the SAP S/4HANA system you need the user `SAP_SATC_ADMIN` to use transaction ATC to perform custom code checks. The user provides the authorizations for setting up and using the ABAP test cockpit (ATC).

You also need the authorization object `S_YCM` for importing the Simplification Database into the system:

Overview of the Authorization Object

Name of the Authorization Object	Names of the Authorization Field	Values of the Authorization Field
S_YCM	SYCM_AREA	SBD
	ACTVT	UL

Applying SAP Notes

We recommend applying the SAP Notes listed in SAP Note 2436688.

Implementing the Custom Code Migration App

Before you can implement the Custom Code Migration app, make sure that you have done the steps for *Implementing SAP Fiori Apps for Administrative, Configuration, and Extensibility Task*, see https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance (English)* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *SAP S/4HANA* ▶ *Enterprise Technology* ▶ *ABAP Platform* ▶ *Configuring the ABAP Platform*.

You need to do settings in both the front-end and the back-end server:

1. **Front-end server:**
    1. a. Enabling the app for access in the SAP Fiori launchpad:
        1. In transaction `PFCG`, add the business catalog `SAP_BASIS_BC_CCM` (Custom Code Migration) to the user roles that need to use the Custom Code Migration app. The catalog can be added to either an existing user role or to a new one.
        2. Assign the user roles to all users that need to use the Custom Code Migration app.
    2. b. Activating OData services
       In transaction `/IWFND/MAINT_SERVICE`, add the following technical service names as services:

       List of Services

Component	External Technical Name
Custom code migration projects	`SYCM_APS_C_PROJECT_CDS`
Analysis of SAP S/4HANA custom code check findings	`SYCM_APS_C_ATC_FIND_ALP_CDS`
Custom code scoping	`SYCM_APS_SCOPING`
Download of SAP S/4HANA custom code check findings	`SYCM_APS_FILE_SRV`

For more information about activating OData services, see: https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance (English)* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *SAP Fiori* ▶ *SAP Fiori Overview* ▶ *Implement SAP Fiori Apps* ▶ *UI-Specific Implementation Tasks* ▶ and http://help.sap.com/fiori_implementation ▶ *Implement* ▶ *SAP Fiori: App Implementation* ▶.

2. **Back-end server: assigning authorizations:**
   Assign the user role `SAP_BC_YCM_APS` to the users who need to work with the Custom Code Migration app. For users who only need to be able to display the results, add the authorization object `S_YCM_APS` to their roles and set the field `ACTVT` to the value *Display*.

Importing the Simplification Database

To check your custom code against the SAP S/4HANA simplifications, you need to download the Simplification Database zip file and import its contents.

SAP provides content for the Simplification Database of an SAP product in the SAP Support Portal. It can be downloaded as a zip file. The content of the zip file must then be imported into the system.

For information about the most recent content of the Simplification Database provided for SAP S/4HANA, see SAP Note 2241080.

1. **Downloading the file:**
   1. Go to the SAP Software Download Center: https://support.sap.com/en/my-support/software-downloads.html.
   2. Choose ▶ *Types of Software* ▶ *Installations & Upgrades* ▶ *Access downloads* ▶ and search for **`CCMSIDB`**.
   3. Select the relevant zip file and choose *Download Basket*.
2. **Importing the content:**
   1. Log on to the SAP S/4HANA system.
   2. Run transaction `SYCM`.
   3. From the menu, choose ▶ *Simplification Database* ▶ *Import from ZIP File* ▶.
   4. Select the zip file with the downloaded Simplification Database from your drive and confirm with *Open*. This imports the Simplification Database content into the system.

3.5.1.2 Using the ABAP Test Cockpit

To check for ABAP test cockpit (ATC) findings, you need to configure a local ATC run series that checks the requested development objects. In addition, you can make use of the baseline functions to filter the findings for the current release (hence not specific to the upgrade itself). With the baseline functions, you can take over individual ATC results into the baseline and so, for example, exclude already known findings from the ATC result list.

To configure ATC run series, schedule them, and add them to the baseline you need to do the following tasks:

1. Configure a local ATC run series for the current release
2. Schedule the ATC run series for the current release
3. Add the ATC run series to the baseline
4. Configure a local ATC run series for the target release
5. Schedule the ATC run series for the target release
6. Analyse the findings

Configure a Local ATC Run Series for the Current Release

You need follow these steps:

1. Log on to your SAP S/4HANA system.
2. Run transaction **ATC**.
   The *ABAP Test Cockpit Overview* screen appears.
3. In the navigation pane, expand the nodes *ATC Administration* > *Runs* and double-click the *Schedule Runs* entry.
   The screen now displays a list of existing run series.
4. Choose the *Create* button in the toolbar.
5. Enter a name for the new series and confirm.
6. Specify the following entries for the new run series:

   Overview of Entries

Fields	Descriptions
Description	Enter a short description for the ATC run series you want to configure. **Note** You can use the template provided by the *Description* field. When you execute the run series, the built-in variables are filled with data for the system, the day of the week, the calendar week, and the year. However, you can also add further texts to these variables, rearrange them, or replace the built-in variables with your own texts.

Fields	Descriptions

Check Variant	Enter one of the following check variants that matches your **current** SAP S/4HANA release:
	`S4HANA_READINESS_<year>_NO_FLE`
	Predefined settings:
	• Material number length = 18
	• Extended amount field length is disabled
	• Extended stock requirement field length is disabled
	• Extended season/theme/collection field length is disabled
	`S4HANA_READINESS_<year>`
	Predefined settings:
	• Material number length = 40
	• Extended amount field length is enabled
	• Extended stock requirement field length is enabled
	• Extended season/theme/collection field length is enabled
	Note
	These product-specific global check variants check all the simplifications relevant for your current SAP S/4HANA release as well as those for previous releases.

Objects to Check	Specify the set of development objects you want to check. On the *Checkable Namespaces* tab, the option *By Query* lets you specify objects by name (such as by package name), by transport layer, or by component. The option *By Object Set* lets you specify an object set that you have defined in Code Inspector in the checked system. **Note** You can use the value help to choose the packages or the object set in the remote system. On the *Modified Objects* tab, you can specify the modified source code objects you want to check.

7. Save the configuration.

Schedule the ATC Run Series for the Current Release

Proceed as follows to schedule a run series:

1. Select the required run series from the list and and click the *Schedule* button in the toolbar.
2. In the dialog that appears, adapt the settings for the *Life Span* of the series and for *Execution* to your needs. Otherwise, leave these settings unchanged.

   > ⓘ Note
   >
   > Note: The *Life Span* defines how long (in days) the ATC results are kept in the system. After this period, the ATC results are automatically deleted in the system.

3. Choose *Execute in Background* (F9) to schedule the run series.

Add the ATC Run Series to the Baseline

Proceed as follows:

1. Run transaction ATC.
   The *ABAP Test Cockpit Overview* screen appears.
2. In the navigation pane, expand the nodes *ATC Administration* > *Runs* and double-click the *Manage Results* entry.
3. In the dialog, adapt the settings to display the last executed ATC run series. Usually, you can keep the default settings.
4. Choose the *Execute* button in the toolbar.
   The screen displays a list of executed run series.
5. Select the relevant row for the ATC run series scheduled for the current SAP S/4HANA release. In the menu, go to *Baseline* > *Add to Baseline to ...* > *Exempt Findings*.
6. Confirm the popup informing you about the addition of the check result to the baseline (select *Yes*).
7. The flag in the column *Is in Baseline* shows that the run series has been added to the baseline.

Configure a Local ATC Run Series for the Target Release

Proceed as described in the section *Configure a Local ATC Run Series for the Current Release* (see above) and specify the following entries for the new run series:

Overview of Entries

Fields	Descriptions
Description	Enter a short description for the ATC run series you want to configure. **Note** You can use the template provided by the *Description* field. When you execute the run series, the built-in variables are filled with data for the system, the day of the week, the calendar week, and the year. However, you can also add further texts to these variables, rearrange them, or replace the built-in variables with your own texts.
Consider Baseline	Flag this field. **Note** When this setting is activated, all findings which are also stored in the baseline are - depending on the baseline option - suppressed, treated as exempted, or will be shown with reduced priority. Note:

Fields	Descriptions

Check Variant	Enter one of the following check variants that matches your **target** SAP S/4HANA release:
	`S4HANA_READINESS_2022_NO_FLE`
	Predefined settings:
	• Material number length = 18
	• Extended amount field length is disabled
	• Extended stock requirement field length is disabled
	• Extended season/theme/collection field length is disabled
	`S4HANA_READINESS_2022`
	Predefined settings:
	• Material number length = 40
	• Extended amount field length is enabled
	• Extended stock requirement field length is enabled
	• Extended season/theme/collection field length is enabled
	Note
	These product-specific global check variants check all the simplifications relevant for SAP S/4HANA 2022, including those for previous releases.

Objects to Check	Specify the set of development objects you want to check. On the *Checkable Namespaces* tab, the option *By Query* lets you specify objects by name (such as by package name), by transport layer, or by component. The option *By Object Set* lets you specify an object set that you have defined in Code Inspector in the checked system. **Note** You can use the value help to choose the packages, or the object set in the remote system. On the *Modified Objects* tab, you can specify the modified source code objects you want to check.

When you have entered your settings, save the configuration.

Schedule the ATC Run Series for the Target Release

Proceed as described in the section *Schedule the ATC Run Series for the Current Release* (see above) and select the relevant run series for the target release. Choose *Execute in Background* (F9) to schedule the run series.

Analyze the Findings

To analyze the results of the ATC run series, proceed as follows:

1. Run transaction `ATC`.
   The *ABAP Test Cockpit Overview* screen appears.
2. In the navigation pane, expand the nodes *ATC Administration* > *Runs* and double-click the *Manage Results* entry.
3. In the dialog, adapt the settings to display the last executed ATC run series. Usually, you can keep the default settings.
4. 4. Choose the `Execute` button in the toolbar.
   The screen displays a list of executed run series.
5. Select the relevant runs series and choose the button *Display* in the toolbar.
   In the dialog, the list of findings is displayed.

> **Note**
>
> In the toolbar, a filter button can be used to display either the unresolved findings (by default) and/or the exempted findings – that, for example, have been included in the baseline.

3.5.1.3 Using the Custom Code Migration App

The Custom Code Migration app (Fiori ID: F3191) enables you to scope and analyse custom code. In addition, the app supports you with identifying unused custom code based on your collected usage data. It provides a graphical representation of custom code analysis results. The results can be filtered by various categories, such as quick fix availability, scope and usage data

To use the Custom Code Migration app to analyse your custom code, you need to do the following tasks:

1. Create a custom code migration project for the current release
2. Add the ATC run series to the baseline
3. Create a custom code migration project for the target release
4. Analyze the findings

Create a Custom Code Migration Project for the Current Release

To create a project, proceed as follows:

1. Start the tile *Custom Code Migration* from the SAP Fiori launchpad.
2. Choose *Create* to create a new custom code migration project.
3. Enter a *Project Description* for your new project.
4. In the field *Target Release* specify your **current** SAP S/4HANA release.

   > **Note**
   >
   > Make sure that the *Material Number Length* is set to the relevant value and the checkboxes for *Extended Amount Length*, *Extended Stock/Requirement Segment* and *Extended Season/Theme/Collection* are enabled, if you use them.

5. To define the *Destination*, choose the communication scenario.
6. Confirm your input with *Save*.

Add the ATC Run Series to the Baseline

Proceed as explained in *Add the ATC Run Series to the Baseline* in section Using the ABAP Test Cockpit [page 22].

Create a Custom Code Migration Project for the Target Release

To create the project, proceed as explained in *Create a Custom Code Migration Project for the Current Release* (see above), except that in the field *Target Release* you specify your **target** SAP S/4HANA release (for example, SAP S/4HANA 2022).

Analyze the Findings

When you create a custom code migration project, the SAP S/4HANA custom code checks are performed automatically. The Custom Code Migration app then gives you an analytical representation of SAP S/4HANA custom code check findings.

To analyze the findings, proceed as follows:

1. After the first analysis has been finished, choose *Analysis* to get an overview of the results.
2. Choose *Analyze Findings* to get a detailed analysis.
3. In the *Custom Code Migration - Analysis - Findings* view, you can specify various filters, for example:
   - *Simplification Item Category* specifies whether findings are related to functionality which is not available any more or has been changed in an incompatible way.
   - *SAP Note Number* lets you analyze findings of a specific simplification Item (for example field length extension of material number).

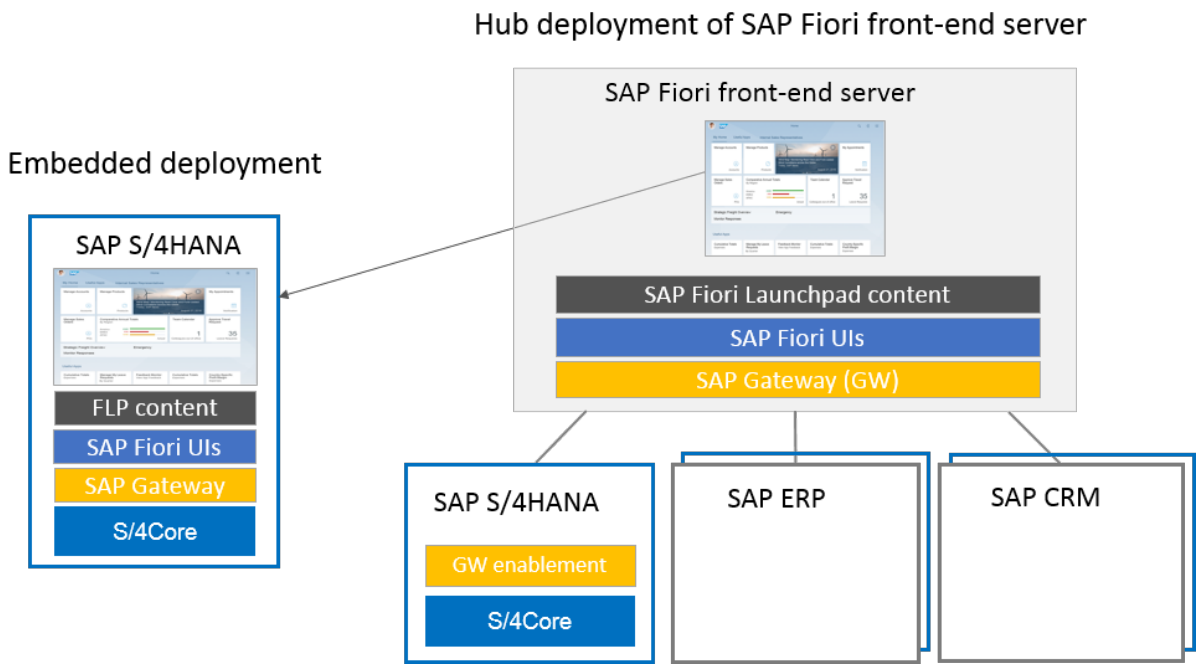
- *Scope Information* lets you filter the findings to show only findings for objects that are in scope.
- *Quick Fix Availability* allows you to analyze which findings can be solved by a Quick Fix and which findings have to be solved manually.

4. Change the settings for the chart.
   To show the quick fix availability per SAP Note title, you must specify the following settings for the chart:
   - SAP Note title (as Category)
   - Findings (as Axis 1)
   - Quick Fix Availability (as Series)

3.6 Upgrade Information for SAP Fiori for SAP S/4HANA

> **Note**
>
> For an overview of the end-to-end approach of an SAP Fiori Upgrade and to support your planning, see https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance* ▶ *English* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *SAP Fiori* ▶ *SAP Fiori Overview* ▶ *Update, Upgrade, or Migrate* ▶.

SAP Fiori can be deployed in an embedded and/or a hub scenario. In the sections below, you find information about the components and versions required for each of the two scenarios.



For more information, see:

- SAP Note 2590653 (about SAP Fiori deployment options)
- *SAP Fiori for SAP S/4HANA 2023: Release Information Note* 3336823

See also Follow-On Activities for SAP Fiori [page 36].

Embedded Deployment

The columns in the table below show the embedded components for the SAP S/4HANA releases 2020 to 2023.

Embedded Deployment

Embedded 2020	Embedded 2021	Embedded 2022	Embedded 2023
Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)
SAP Fiori for SAP S/4HANA 2020	SAP Fiori for SAP S/4HANA 2021	SAP Fiori for SAP S/4HANA 2022	SAP Fiori for SAP S/4HANA 2023
SAP FES 2020 for SAP S/4HANA	SAP FES 2021 for SAP S/4HANA	SAP FES 2022 for SAP S/4HANA	SAP FES 2023 for SAP S/4HANA
SAP_UI 7.55	SAP_UI 7.56	SAP_UI 7.57	SAP_UI 7.58
Or SAP FES 2021	**Or SAP FES 2022**		
Or SAP FES 2022			
SAP S/4HANA 2020	SAP S/4HANA 2021	SAP S/4HANA 2022	SAP S/4HANA 2023
SAP Gateway 7.55	SAP Gateway 7.56	SAP Gateway 7.57	SAP Gateway 7.58
ABAP Platform 2020	ABAP Platform 2021	ABAP Platform 2022	ABAP Platform 2023
See also, SAP Note 2935725	See also, SAP Note 3085560	See also, SAP Note 3237135	See also, SAP Note 3348673

Standalone Deployment

The columns in the table below show the standalone and backend components for the SAP S/4HANA releases 2020 to 2023.

Standalone Deployment

Standalone 2020	Standalone 2021	Standalone 2022	Standalone 2023
Frontend	**Frontend**	**Frontend**	**Frontend**
Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)	Optional UI Components (HR, Travel, MDG)
SAP Fiori for SAP S/4HANA 2020	SAP Fiori for SAP S/4HANA 2021	SAP Fiori for SAP S/4HANA 2022	SAP Fiori for SAP S/4HANA 2023

Standalone 2020	Standalone 2021	Standalone 2022	Standalone 2023
SAP FES 2020 for SAP S/4HANA	SAP FES 2021 for SAP S/4HANA	SAP FES 2022 for SAP S/4HANA	SAP FES 2023 for SAP S/4HANA
SAPUI5 1.78	SAPUI5 1.90	SAPUI5 1.102	SAPUI5 1.114
SAP Gateway 7.52 (minimum version)	SAP Gateway 7.52 (minimum version)	SAP Gateway 7.52 (minimum version)	
SAP NetWeaver 7.52 SP07 (minimum version)	SAP NetWeaver 7.52 SP08 (minimum version)	SAP NetWeaver 7.52 SP11 (minimum version)	
Backend	**Backend**	**Backend**	**Backend**
SAP S/4HANA 2020	SAP S/4HANA 2021	SAP S/4HANA 2022	SAP S/4HANA 2023
SAP Gateway 7.55	SAP Gateway 7.56	SAP Gateway 7.57	SAP Gateway 7.58
ABAP Platform 2020	ABAP Platform 2021	ABAP Platform 2022	ABAP Platform 2023
See also, SAP Note 2935725	See also, SAP Note 3085560	See also, SAP Note 3237135	See also, SAP Note 3348673

Recommended SAP Fiori Apps

You can use the SAP Readiness Check for SAP S/4HANA upgrades to recommend SAP Fiori apps based on your transaction usage history, see the section *Recommended SAP Fiori Apps* at https://sap.com/readinesscheck ▶ *Feature Scope Description* > *SAP Readiness Check* > *Functions of SAP Readiness Check for SAP S/4HANA Upgrades*.

3.7 Upgrade Information for Java Instances

If you plan an upgrade for SAP S/4HANA Java instances to SAP S/4HANA 2023 (any feature package or support package stack), first check that your operating system is still supported. For more information, see the Product Availability Matrix at https://apps.support.sap.com/sap/support/pam.

If your operating system is no longer supported, you have to migrate to a supported operating system using the system copy procedure. For information see the system copy guide at https://support.sap.com/sltoolset ▶ *System Provisioning* > *System Provisioning Scenarios* > *Copy a System using Software Provisioning Manager*.

3.8 Upgrade Information for Operational Data Provisioning for SAP BW

When upgrading to SAP S/4HANA 2023, software component SAP BASIS is also updated to Version 7.58. With the upgrade of SAP BASIS 7.53, two technical job definitions automatically became active:

- `SAP_ODQ_CLEANUP`: Scheduling of program `ODQ_CLEANUP` for reorganization of delta queue
- `SAP_ODQ_DAEMON_STARTER`: Scheduling of program `ODQ_DAEMON` for support of real-time subscriptions

Before the upgrade, please check whether you have scheduled program `ODQ_CLEANUP` or `ODQ_DAEMON` with your own variants. In this case, you must decide whether you continue with the scheduling of your own variants after the upgrade, or instead, want to use the variants recommended by SAP in the technical job definitions. If your own variants correspond to the variants recommended by SAP, choose to use the technical job definitions.

For more information, see SAP Note 2677355.

3.9 Data Transition Validation

The data transition validation is a tool that allows you to compare business data before and after a system conversion from SAP ERP to SAP S/4HANA, or before and after an SAP S/4HANA upgrade. The tool supports validation of upgrade or conversion scenarios with predelivered content for business validation. The tool uses business reports or transactions as instruments of comparison. It is available as part of the standard license.

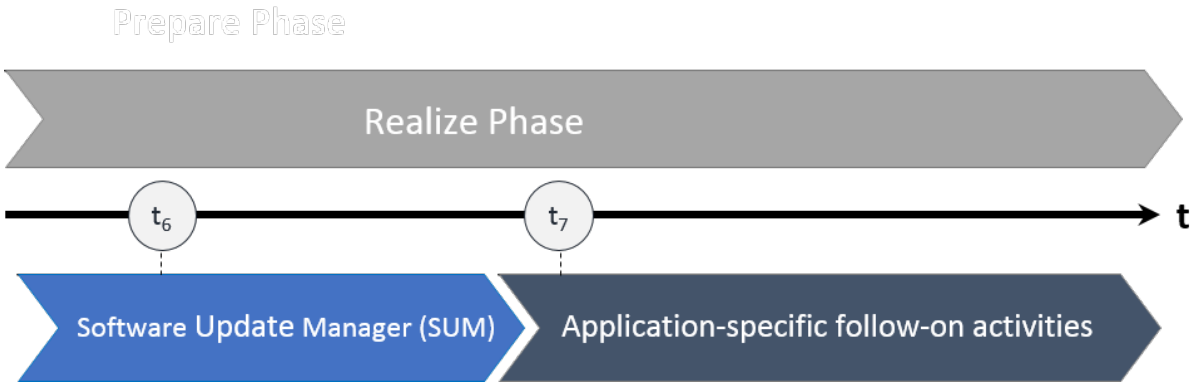
To use DTV for the business validation after an upgrade, proceed as follows:

1. In the preparation phase of the uprgade, define the reports to be used for validation together with your requesting lines of business and your internal or external auditors.
2. Create a project and maintain required test specifications in the DTV tool.
3. During the ramp-down of the system prior to the SAP S/4HANA system upgrade, trigger data extraction from the source release.
4. Once the SAP S/4HANA system upgrade is finished, proceed with the tasks in the target release to validate business correctness.

For more information about the DTV tool, see:

- Product documentation at https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance* ▶ *English* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *ABAP Platform* ▶ *Administrating the ABAP Platform* ▶ *Administration Concepts and Tools* ▶ *Solution Life Cycle Management* ▶ *Software Logistics* ▶ *Data Transition Validation*
- SAP Note 3117879

4 Realizing the Upgrade



Upgrade to SAP S/4HANA: Realization

After the preparation phase, you start the upgrade using the Software Update Manager (SUM) tool. Make sure that you also check the upgrade-relevant simplification items! For a short overview of the entire process, see Overview of the Upgrade Process [page 8].

Software Update Manager (SUM) is the technical tool used for the upgrade to SAP S/4HANA.

> ⓘ Note
>
> You have to run the Maintenance Planner before the SUM, because the SUM requires the `stack.xml` generated by the Maintenance Planner as an input. For more information, see Maintenance Planner [page 15].

For all information about using the SUM, see the guide *Software Update Manager 2.0 <latest version>* available at https://support.sap.com/sltoolset *Software Logistics Toolset* *System Maintenance* *System Maintenance Scenarios* *Software Update/Upgrade using SUM*.

> ⓘ Note
>
> Simplification Item-Check and custom-code migration checks are carried out by the SUM tool. It will provide you with a list of results that may include manual application-specific activities you need to do before the upgrade can be finalized.
>
> Please see also SAP Note 3340526.

4.1 Silent Data Migration

Migration of application data usually happens during release upgrades or updates during downtime. In contrast, the Silent Data Migration Infrastructure (SDMI) allows zero-downtime upgrades and runs after the upgrade (or conversion) during business operations.

An SDMI supports uptime capability for at least one release. Therefore zero-downtime upgrades are possible if every release is taken into account. If an upgrade jumps over a certain release (for example, if the upgrade goes from SAP S/4HANA 1809 straight to SAP S/4HANA 2020), the SDMIs from the release that was jumped over (in this example, SAP S/4HANA 1909) are done during business downtime (if they do not support uptime capability for the target release).

As of SAP S/4HANA 2020 (as start release) SDMIs also support **three** releases in uptime making it possible to do jump-upgrades with zero downtime option (ZDO) over up to **two** releases (towards the **third** release), for example from SAP S/4HANA 2020 to SAP S/4HANA 2023.

By default the SDMIs are run during uptime. The SDMI job from the job repository starts and calls the SDMIs in each client individually once the system comes out of the maintenance mode. To see all relevant SDMIs, call transaction `SDM_MON`.

The users can be maintained in transaction `SDM_USER` prior to the upgrade, so that the users can comply to your corporate standard. However, they can also be created as technical users during the upgrade process by the Software Update Manager (SUM). During the post-processing phase, the SUM checks if the technical user exists. If not, a message will be created in the update log. If the users are not created, the SDMIs cannot be started and the data will not be migrated.

> **Note**
>
> It is also possible to create the technical users after the SUM run. For more information, see https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance* ▶ *English* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *ABAP Platform* ▶ *Adminstrating the ABAP Platform* ▶ *Administration Concepts and Tools* ▶ *Solution Life Cycle Management* ▶ *Silent Data Migration Infrastructure* ▶.

The SUM offers the option to force the SDMI processing during downtime. In that case, the SDMIs are finished by the time the system resumes business operations.

> **Caution**
>
> On the one hand, this option has the one benefit that the business data is no longer changed by SDMI processes after the upgrade procedure is completed. On the other hand, this option also includes disadvantages that you should be aware of:
>
> - The overall system business downtime is significantly extended due to the time it takes to execute the data migration because the SUM runs the SDMI jobs synchronously and waits until they are completed.
> - Depending on the available resources, the size of the migration tables and the configuration of the dialog work processes and the parallelization framework, the downtime can even reach a duration that is no longer tolerable.

> **Note**
>
> Please take care that all SDMIs have been processed successfully in all clients. Otherwise this will leave the upgrade unfinished and bears the risk, that some new application functionality may not be available on the target release. Furthermore, all SDMIs need to be done before the next upgrade can happen.

For more information, see:

- SAP Note 2850918 *Authorizations required for Silent Data Migration (SDMI) in SAP S/4HANA Release*

- SAP Note 2664638 *Create and assign SDMI User in a client*
- SAP Note 2907976 *Silent Data Migration (SDMI) - FAQ*
- SAP Note 2916801 *Silent Data Migration (SDMI) Configuration Options*
- SAP Note 3382097 *SDMI Classes in the Silent Data Migration Framework in SAP S/4HANA 2023 (Details, Corrections, Special Handling)*
- SAP Note 3445802 *Silent Data Migration (SDMI) New Rows after Migration - FAQ*
- https://blogs.sap.com/2019/10/03/s4hana-silent-data-migration-infrastructure-quick-guide/

4.2 Cleaning Up Obsolete Data After the Upgrade

The Obsolete Data Handling tool allows you to delete obsolete data that may remain after the upgrade.

Obsolete data originates from SAP S/4HANA data model simplifications in many application areas where the related application data is migrated to new data structures. These areas include Material Management, Financials, Sales and Distribution, Environment and Health Science and others. The source data is not deleted automatically during the conversion because this would increase the business downtime and the obsolete data store may sometimes be required during or after the upgrade. Therefore, after you have successfully tested and validated the upgrade results, you can clean up the obsolete data with the Obsolete Data Handling tool.

> **Note**
>
> With SAP S/4HANA 2023 a new obsolete data handling tool is delivered.
>
> The old tool (report/transaction code `ODH_Data_processing`) can only be used with SAP S/4HANA 2023 to view the logs of already finished cleaning runs. It will not be available at all for future SAP S/4HANA releases. For the old tool, see SAP Note 2661837.
>
> **All new obsolete data cleaning activity must be done with the new, class-based tool** (transaction code `SODH`). For information on how to enable and use this tool in the productive SAP S/4HANA 2023 system, see SAP Note 3335020.

4.3 Follow-On Activities for SAP Fiori

Before doing any of the below steps, make sure that you have checked the information about the SAP Fiori upgrade at https://help.sap.com/s4hana_op_2023 *Use* > *Product Assistance* > *English* > *Related Information* > *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* > *Enterprise Technology* > *SAP Fiori* > *SAP Fiori Overview* > *Update, Upgrade, or Migrate* > *Upgrade Support of SAP Fiori for SAP S/4HANA* > *Follow-on Activities after SAP Fiori Upgrade* >.

> **Note**
>
> If you use Rapid Activation to automate the activation of SAP Fiori for SAP S/4HANA, some tasks might already have been processed.

For details about the Rapid Activation concept for SAP Fiori in SAP S/4HANA, see SAP Note 2902673 and the section *Rapid Activation of SAP Fiori* at https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance* ▶ *English* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *SAP Fiori* ▶ *SAP Fiori Overview* ▶ (use the search to find the section).

Please perform the following tasks on your **front-end server** after the technical upgrade with the Software Update Manager (SUM) to SAP S/4HANA 2023 to ensure that your SAP Fiori Launchpad and SAP Fiori Applications are able to run properly:

1. Check the latest correction notes from the *SAP Fiori for SAP S/4HANA 2023 Release Information Note* 3336823.
2. Check that the latest corrections for task lists needed to activate and update SAP Fiori content have been applied. These task lists are often needed to resolve obsolete/deprecated content, and for post-upgrade activities. All SAP Notes for your target SAP S/4HANA release and FPS are listed in SAP Note 2902673.
3. Replicate app descriptors:
   If you are still using untyped replicable catalog types (BEC) (also known as backend catalogs for classic UIs) in your SAP S/4HANA source system, you need to replicate backend catalogs by running the report `/UI2/GET_APP_DESCR_REMOTE_ALL` on your front-end server to start the replication.

   > **Note**
   >
   > This replication step is also required in an embedded deployment scenario.

4. Invalidate server caches to apply configuration changes:
   Run the reports `/UI2/INVALIDATE_CLIENT_CACHES` and `/UI2/INVALIDATE_GLOBAL_CACHES`.

> **Note**
>
> The above steps are also required when updating to a new service pack (SP) or feature package stack (FPS).

Please do the following task on your **back-end server**:

If you use SAP GUI for HTML Apps on your SAP Fiori Launchpad and there are issues after the upgrade, check if the ICF node `/sap/public/bc/its` and all its sub-nodes are activated (use transaction `SICF`). To resolve ICF services inactivated by the upgrade, run report `RS_ICF_SERV_MASS_PROCESSING`.

SAP Fiori Launchpad Settings for Improved Performance

We recommend that you review the end-to-end SAP Fiori performance after the upgrade, see SAP Note 2916959.

For more information on how to configure the SAP Fiori Launchpad, see https://help.sap.com/s4hana_op_2023 ▶ *Use* ▶ *Product Assistance* ▶ *English* ▶ *Related Information* ▶ *SAP S/4HANA and SAP S/4HANA Cloud Private Edition* ▶ *Enterprise Technology* ▶ *ABAP Platform* ▶ *UI Technologies* ▶ *SAP Fiori Launchpad* ▶.

4.4 Default Authorization Values and PFCG Roles

> **Note**
>
> Please make sure that you have first done the steps and checked the information in the section Follow-On Activities for SAP Fiori [page 36].

Checking Default Authorization Values After the Upgrade

After the upgrade, you must do some follow-up activities regarding the authorization concept.

The default authorization values are the data basis for the provision of authorizations required for applications in the Profile Generator (transaction **PFCG**).

Default authorization values for standard SAP applications are delivered via upgrade, support package deployment, or SAP Note. This SAP-provided data is displayed and maintained in transaction **SU22**.

Customer default authorization values are maintained in transaction **SU24**. Only these customer default authorization values are applied when PFCG is used.

After the upgrade the SAP default values may have been changed. You therefore must use transaction **SU25** to carry out a comparison of the default values (steps 2a and 2b) and transfer the default values maintained by SAP.

For more information about this process, see SAP Notes 1539556 and 440231.

Updating PFCG Roles

After checking and updating the default authorization values you must do some follow-up activities for the PFCG roles. You need to consider how changed default authorization values as well as new or changed content in SAP Fiori Catalogs affect your PFCG roles. You adjust your PFCG roles using the transaction **SU25** actions *Update of Application Groups in Role Menu* and *Roles to Be Checked*.

4.5 Follow-On Activities for Advanced Variant Configuration

- Tasks required after the upgrade from SAP S/4HANA releases lower than 1709:
  In this case, you have to all the steps described in the section *Follow-Up Activities for Advanced Variant Configuration* in the *Installation Guide for SAP S/4HANA* at https://help.sap.com/s4hana_op_2023 ▶ *Implement* ▶ *Guides* ▶.
- Tasks required after the upgrade from SAP S/4HANA releases equal to or higher than 1709:
  In this case, you have to do the steps described in SAP Note 2987015.

4.6 Follow-On Activities for PP/DS

Handling of material plants that are *Flagged for Deletion* in PP/DS has been changed as of SAP S/4HANA 2022. If you are upgrading from SAP S/4HANA 2021 or lower to SAP S/4HANA 2022 or higher, you must run the report `PPDS_DEACTIVATE_ADVN_PLN_PROD` manually to update the *Advanced Planning Status* for material plants that are *Flagged for Deletion*. However, if you are upgrading from SAP S/4HANA 2022 to SAP S/4HANA 2023, you do not need to run the report `PPDS_DEACTIVATE_ADVN_PLN_PROD` manually because the data in the SAP S/4HANA 2022 system is already correct.

> **Note**
>
> For more information, see SAP Note 3150247.

4.7 Follow-On Activities for Policy Management (FS-PM)

With Policy Management (FS-PM), you can map the whole life cycle of an insurance contract, starting from the creation of an application, through policy issuance and ongoing contract maintenance, right up to the termination of the contract.

SAP Notes for the Upgrade of Policy Management

SAP Note Number	Title
3373830	Release Information Note (RIN) for Policy Management and Claims Management in SAP S/4HANA 2023

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